

REPORT ON GRID NAVIGATION ALGORITHMS: VALUE ITERATION VS. POLICY ITERATION

1. Introduction

This report presents the findings from an analysis of two computer science notebooks. The objective of these notebooks was to teach a digital agent (like a virtual robot) how to navigate an 18x18 grid environment. To solve this, the notebooks tested two well-known methods: Value Iteration and Policy Iteration. The goal of both methods is to help the agent find the best possible path to a highly rewarding destination while avoiding dead ends or penalties.

2. How We Measure Success

Instead of using complicated mathematics, we can understand the grid as a simple map. Every single square on this 18x18 map gets a "utility" score. A high score means the square is a great place to be (close to the goal), and a low score means it is a bad place to be (like a trap). The two algorithms try to calculate the perfect score for all 324 squares on the grid so the agent always knows which direction is safest and most rewarding.

3. Findings on Value Iteration

The first notebook focused on Value Iteration. This method works by making a blind guess about the value of each square and then slowly updating those guesses based on the squares around it.

- **Slow but Steady Progress:** Because it only updates values one tiny step at a time, it takes a long time to finish. The data logs showed that it had to repeat its calculations hundreds of times (up to iteration 693) before it finally figured out the exact map.
- **Simple Steps:** Even though it took many steps, the math required for each individual step was very simple and fast for the computer to process.

4. Findings on Policy Iteration

The second notebook focused on Policy Iteration, which takes a very different approach. Instead of just guessing scores, this algorithm creates a strict set of rules (called a "policy") that tells the agent exactly which way to step on every square. It then grades how good those rules are and improves them over time.

- **Fewer Total Steps:** Policy Iteration is much more direct. The logs indicate that it requires far fewer total loops to find the perfect solution compared to Value Iteration.
- **Heavy Calculations:** The trade-off is that evaluating and changing the "rules" takes a lot more computing power per step. However, it gets to the final answer much quicker in terms of overall iterations.

5. Visual Insights and Convergence

When we look at the data at the end of the notebooks, two things become very clear:

- **The Convergence Point:** When we track the changes happening inside the grid, we see large adjustments in the beginning. Eventually, the changes drop to zero (stabilizing). This proves that the algorithms successfully learned the map and stopped calculating once they found the optimal path.
- **The Final Heatmap:** Extracting the final scores and putting them back into the 18x18 grid creates a clear visual map. The final outputs clearly highlight the most valuable squares near the goal and correctly mark the dangerous zones with low scores, proving the algorithms worked exactly as intended.

6. Conclusion

Both Value Iteration and Policy Iteration successfully solved the grid navigation problem and arrived at the exact same correct answers. The main difference lies in their approach. Value Iteration takes many simple, repetitive steps to slowly spread the right values across the map. Policy Iteration takes fewer, but heavier, steps by actively improving its driving rules. For this specific 18x18 environment, both are highly effective, but Policy Iteration proves to be a more direct method for finding the final navigation rules.